

AI Engineer Trial Assignment

System Architecture Document

Architecture

Arxiv API

↓

Research Paper Collection

↓

Gemini LLM Extraction

↓

Entity Resolution

↓

CSV Storage

↓

GitHub Repository

Components

1. Data Collection Layer
2. LLM Processing Layer
3. Entity Resolution Layer
4. Output Layer

Error Handling

- Retry logic
- API failure handling
- Entity matching thresholds

Scalability

The architecture can be scaled by:

- Async scraping
- Queue systems
- Distributed workers